

Eshani Patel

Software Engineer - Systems & Infrastructure, LinkedIn, Mountain View, California, USA

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EDUCATION

California Institute of Technology, Pasadena, California, USA

SEP 2021 — MAR 2025

B.S. IN COMPUTER SCIENCE (MAJOR) AND INFORMATION AND DATA SCIENCE (MINOR)

- Relevant Courses: AI/ML Learning Systems, Large Language/Vision Models, Machine Learning/Data Mining, LLM Agents, Relational Databases, Networks, Web Development, Computer Systems, Algorithms, Data Structures, Software Design, Applied Linear Algebra and Probability/Statistics, Statistical Inference, Discrete Math, Distributed Computing.
- GPA: 4.1/4.3

RESEARCH EXPERIENCE

Yue Group, Caltech

Pasadena, California

Guest and Undergrad Researcher

MAR 2025 — Present

- Designed a novel training paradigm extending the Population Transformer, a self-supervised foundation model for intracranial EEG (iEEG), to handle variable-length neural sequences via data-augmented pretraining, enabling population-level representation learning across input lengths.
- Optimized training pipeline memory efficiency by 30%, enabling large-scale model training on smaller-spec servers.
- First-authored paper accepted at NeurIPS 2025 (Foundation Models for the Brain and Body Workshop).

Computational Cameras Group, Caltech

Pasadena, California

Undergrad Researcher

JAN 2024 — MAR 2025

- Investigated scalability and efficiency of Image Generation Models (IGM) method for high-resolution images using StyleGAN to address limitations in denoising and reconstruction tasks.
- Developed a multimodal conditional diffusion model integrating simulated astronomical observational modalities for dark matter field reconstruction and experimentally demonstrated robustness across SNR levels. Submitted work-in-progress for non-archival review at AI2ASE workshop.

Science Understanding through Data Science, NASA Jet Propulsion Laboratory

Pasadena, California

Machine Learning Intern

JUN 2022 — AUG 2022

- Built Random Forest models to predict atmospheric ozone level and bias between physical model and real world data.
- Provided key insight on drivers of ozone and bias identify physical drivers using permutation feature importance.
- Integrated Google Earth Engine-derived topographic data via feature engineering, improving predictive accuracy by 15% and informing physical model bias correction.

PROFESSIONAL EXPERIENCE

Core Infra, LinkedIn

Sunnyvale, California

Software Engineer, Systems and Infrastructure

AUG 2025 — Present

- Building exabyte-scale multi-tenant distributed file and blob storage system to support LinkedIn's data infrastructure.
- Worked on low-latency distributed key-value store for metadata services across LinkedIn's infrastructure (200K QPS).

Data and AI Platform, LinkedIn

Sunnyvale, California

Software Engineering Intern

JUN 2024 — SEP 2024

- Developed and deployed XGBoost-based predictive models to estimate compute usage for on-demand report generation on the experimentation platform.
- Integrated models via a Java microservice API, serving multiple use cases across experimentation platform.
- Orchestrated ETL pipelines with Apache Spark (Scala) to process 1M+ usage records.

Catalog Platform, Rakuten

San Mateo, California

Software Engineering Intern

JUN 2023 — AUG 2023

- Implemented multi-component product history logging in Java for Catalog Platform big data pipeline, enhancing data traceability and supporting analytics at scale of catalog items.
- Utilized Google Cloud Pub/Sub, Dataflow, Big Query, and Apache Beam for ETL of product information in data pipeline.

PUBLICATIONS

Patel, E., Yue, Y., Chau, G. (2025). Learning Time-Scale Invariant Population-Level Neural Representations *Conference on Neural Information Processing Systems (NeurIPS): Workshop on Foundation Models for the Brain and Body* [Paper]

Patel E., Marchetti Y., Miyazaki K., Bowman K., Lu S., Montgomery J. (2023). Towards Understanding the Sub-scale Drivers of Surface Ozone with Machine Learning Based Feature Permutation Importance Analysis, *American Meteorological Society Conference, Student Poster Session* [Paper]

Miyazaki K., Bowman K., Marchetti Y., Lu S., Montgomery J., Li J., **Patel E.** (2023). Driving Mechanisms of Global Surface Ozone and Its Bias in the Current Chemical Reanalysis Using Machine-Learning Approaches, *American Meteorological Society Meeting* [Paper]

TEACHING EXPERIENCE

Caltech Computer Science

Head Teaching Assistant

Pasadena, California

JAN 2022 — Present

- Managed and led 20-person TA staff as Head TA for Software Design course.
- Taught office hours and labs for 100+ students in *Data Structures & Algorithms (Java)* and *Software Design (C)*, consistently receiving top TA evaluations.
- Developed projects (physics engine and game) to improve curriculum for Software Design course.

Technical Interview Preparation, Caltech CS 12 Course

Course instructor

Pasadena, California

JAN 2023 — Present

- Co-designed and taught a 10-week algorithms & systems interview prep course (50 students/term).
- Interviewed and trained students who went to intern at Google, Meta, Amazon, Microsoft, Apple, and more.
- Hired and trained an undergraduate teaching assistant to assist with course activities and grading.

AWARDS

Michael and Edwenna Werner SURF Fellowship

Awarded a fully funded summer undergraduate research from Caltech at NASA's Jet Propulsion Laboratory for outstanding research in the field of climate science.

JUL 2022

Pasadena, California

Academic All-District Award

Selected by College Sports Communicators to recognize the nation's top student-athletes for their combined performances on the court and in the classroom.

Feb 2023

Pasadena, California

SELECTED PROJECTS

- **Applying Fourier-based Neural Networks to Eikonal Tomography:** Applied Fourier Neural Operators for seismic wave travel time estimation to replace traditional computationally intensive PDE, achieving scalable learning on 100GB datasets using GPU-accelerated HPC clusters.
- **ReAct-Based Semantic Wikipedia Scraper:** combined chain-of-thought + action (*ReAct*) with semantic Wikipedia search, achieving higher retrieval precision over standard RAG baselines.
- **AI Diagnostician:** Fine-tuned large vision models (ResNet, DenseNet, ViT) on CheXpert for chest X-ray classification.
- **Mindful Beauty:** Built a full-stack e-commerce platform (MySQL, JS, HTML/CSS) with comprehensive CRUD functionality and real-time filtering.

SKILLS

- **Programming Languages:** Python, Java, C/C++, Rust, Scala, SQL, JavaScript, OCaml
- **Software:** PyTorch, Scikit-learn, Apache Spark, Apache Beam, Kubernetes, Docker, GRPC, Github, Linux, Google Cloud Platform, Azure, Open AI API, NumPy, Pandas, Matplotlib, MATLAB, Node.js
- **Core ML Expertise:** Foundation Models, Generative Models (Diffusion, GANs, VAEs), Transformers, Representation Learning, Multimodal Learning, Deep Learning.
- **Software Disciplines:** AI/ML, Distributed Systems, Databases, Big Data, Cloud Computing, Backend Engineering.